

UI/UX Design System Specification

Source: docs/UI_UX.md • Status: Verified & Synchronized

Product: Cubyntra

Organization: Necookie Labs

Status: Implemented

Module: src/components/, src/app/

1. Design Philosophy: "Technical Laboratory"

Cubyntra's user interface is conceived as a high-precision optical laboratory instrument: clean, dark, high-contrast, uncluttered, and information-dense without feeling overwhelming.

Guiding Principles:

- Clarity Over Clutter: Every UI element conveys actual system state; no superficial widgets.
- Tactile Feedback: Interactive elements provide immediate visual and audio-tactile response.
- Continuous Spatial Orientation: The user is never left guessing which face to scan next or which rotation to execute.

2. Color Palette & Typography

2.1 Theme Tokens

- Background Root: Obsidian void #030712 (Zinc 950) with subtle animated SVG hexagonal grid pattern.
- Card / Surface: Translucent obsidian with glassmorphic blur #0b0f19 / rgba(15, 23, 42, 0.75).
- Primary Accent: Electric Cyan #06b6d4 (Tailwind Cyan 500) and Neon Teal #14b8a6.
- Success / Stability: Emerald Green #10b981 (Tailwind Emerald 500) indicating locked color stability.
- Warning / Unstable: Amber Orange #f59e0b (Tailwind Amber 500) indicating camera jitter or motion blur.
- Error / Parity Alert: Crimson Rose #f43f5e (Tailwind Rose 500).

2.2 Standard Rubik Facelet Colors

- White (U): #ffffff
- Yellow (D): #eab308
- Green (F): #22c55e
- Blue (B): #3b82f6
- Red (R): #ef4444
- Orange (L): #f97316

2.3 Typography

- Primary UI: Sans-Serif (Geist Sans, Inter, system-ui) at weights 400, 500, 600, 700.
- Technical Readouts & Moves: Monospace (Geist Mono, JetBrains Mono) for move sequences (R U R' U'), coordinates, and confidence metrics.

3. Core User Experience Flow

```
[ Welcome / Idle State ]
  %
  %
[ Camera Initialization ] % % % (Denied) % % % % [ Manual Input / Scramble Mode ]
  % (Granted)
```

```

        %¼
[ Guided 6-Face Scan ] %Ä% % % % % % % % % % % % % % % %
  (U -> R -> F -> D -> L -> B)           % (Rescan)
        %                               %
        %¼                               %
[ State Validation Engine ] % % (Invalid) % % (Error Modal Recovery)
        % (Valid)
        %¼
[ Sub-Second Kociemba Solve ]
        %
        %¼
[ Interactive 3D Playback & Step-by-Step Guidance ]

```

4. Key UI Components & Interactions

4.1 Guided Camera Scanner HUD (CameraScanner.tsx)

- Viewport Reticle: 3x3 rounded bounding grid centered over the live video stream.
- Stability Indicator: A circular progress ring around each sticker cell filling up over 10 consecutive stable frames.
- Color Confidence Bar: Real-time readout of CIELAB distance matching confidence.
- Face Progress Tracker: 6 segmented indicators at the top showing completed vs remaining faces.
- Orientation Guide Widget: Mini 3D preview cue showing how to rotate the physical cube to reach the next required face.

4.2 Mathematical Error Recovery Modal (ErrorRecoveryModal.tsx)

- Triggered automatically if state validation detects parity or count anomalies.
- Presents an unfolded 2D cube net view (U, L, F, R, B, D).
- Highlights problematic pieces with pulsing red outlines.
- Provides an intuitive color palette selector to manually override misclassified stickers.

4.3 Solve Playback Controls (SolveControls.tsx)

- Move Tape: Horizontal scrolling carousel of move tokens (e.g. [U] [R2] [F'] [D]). The current move is highlighted in bright cyan with enlarged typography.
- Playback Controls:
 - Play / Pause (toggleable with Spacebar).
 - Step Forward / Step Backward (Right Arrow / Left Arrow).
 - Speed Adjustment: \$0.5\times\$, \$1\times\$, \$2\times\$, \$4\times\$.
 - Reset to initial scramble (R).
 - Progress Gauge: Real-time percentage indicator and step counter (\$K / N\$).

4.4 Real-Time Computer Vision Debugger (CVDebugger.tsx)

- Collapsible diagnostics drawer for advanced users and developers.
 - Displays live RGB, HSV, CIELAB color values for all 9 grid points.
 - Frame rate monitor (FPS) and frame drop detection.
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5. Responsive Design & Accessibility

- Desktop ($\geq 1024\text{px}$): Dual-pane split view with video HUD/controls on the left and the 3D twin visualizer on the right.
 - Mobile ($< 1024\text{px}$): Vertically stacked layout with sticky playback controls at the bottom viewport boundary.
 - Accessibility:
 - All interactive buttons include descriptive aria-label attributes.
 - High contrast ratio exceeding WCAG AA standards (minimum 4.5:1 for normal text, 7:1 for large display metrics).
 - Keyboard navigation fully supported across all critical playback actions.
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